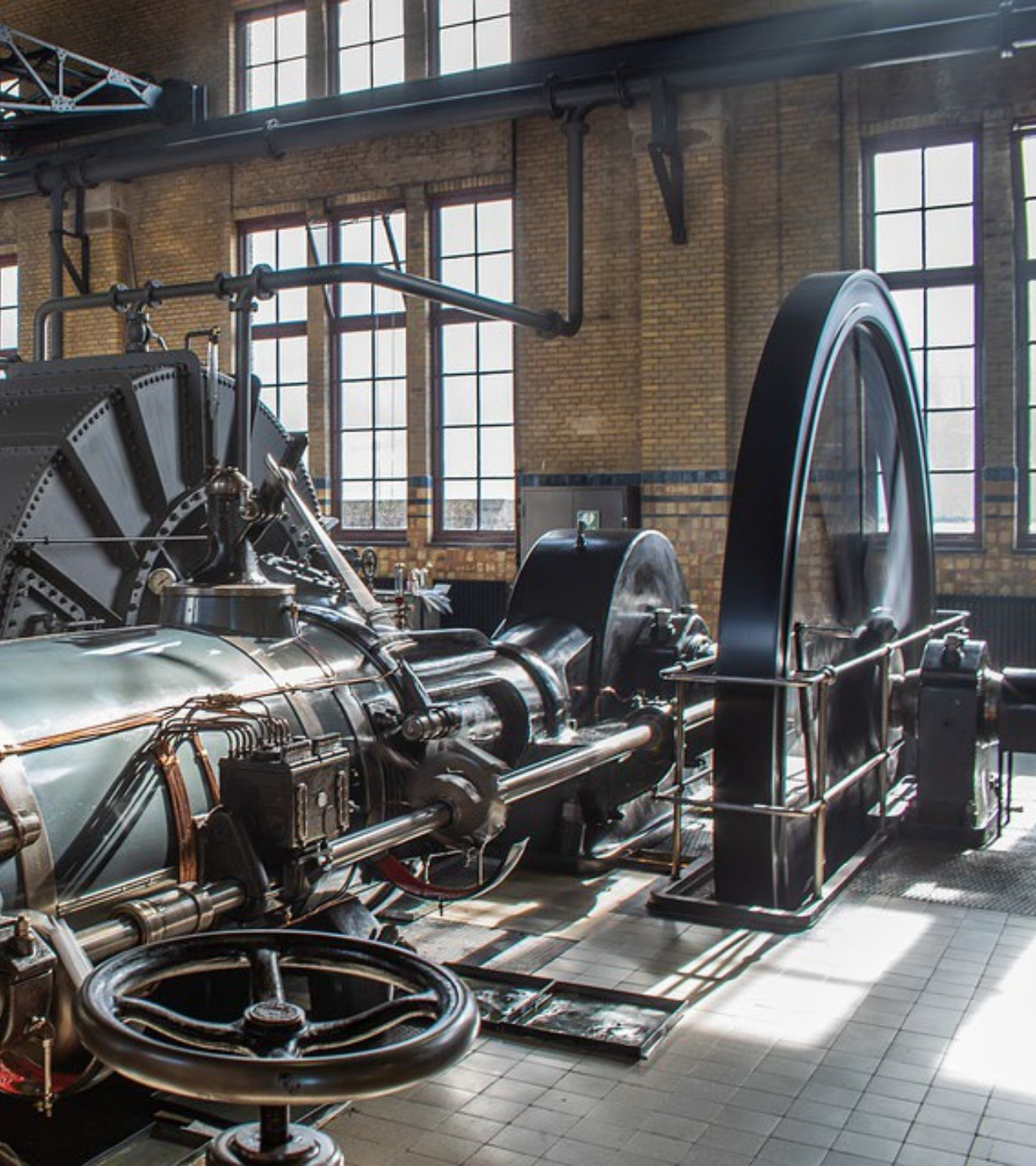




# ME 300: Engineering Thermodynamics

## Example: Stirling Cycle

# Heat engine – example

1-2:  $\Delta T = 0$  compression

2-3:  $\Delta V = 0$  heat in

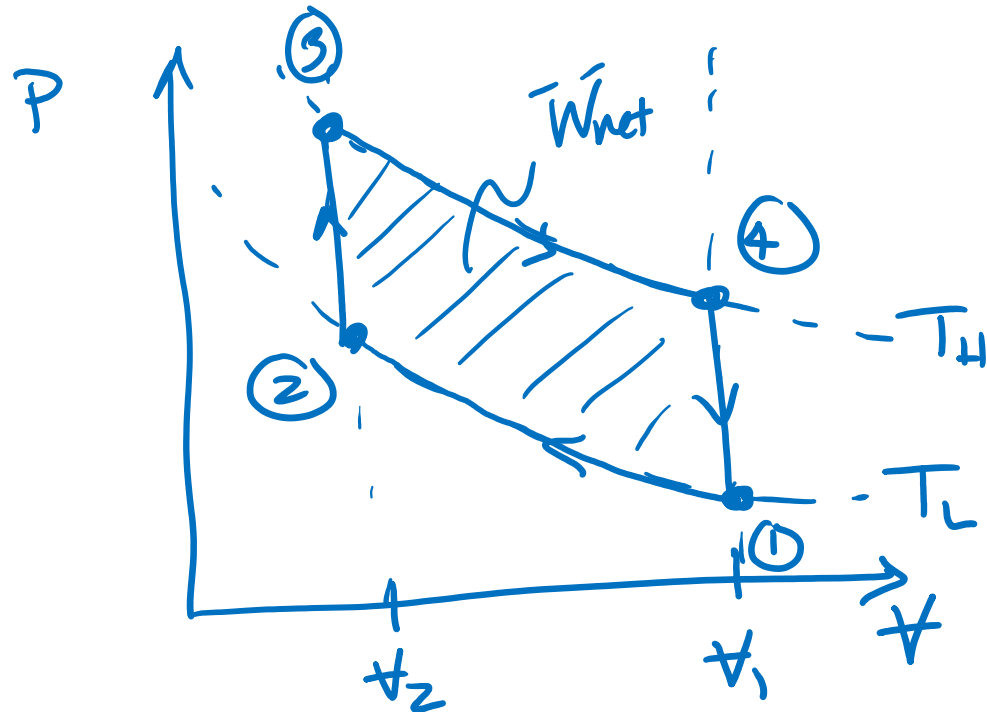
3-4:  $\Delta T = 0$  expansion

4-1:  $\Delta V = 0$  heat out

## Goal

1. State point
2.  $\bar{W}, Q$
3.  $\eta_{th}$

| <u>State</u> | <u>T</u> | <u>V</u> | <u>P</u> if ideal gas     |
|--------------|----------|----------|---------------------------|
| 1            | $T_L$    | $V_1$    | $P_1 = \frac{NkT_1}{V_1}$ |
| 2            | $T_L$    | $V_2$    | $P_2$                     |
| 3            | $T_H$    | $V_2$    | $P_3$                     |
| 4            | $T_H$    | $V_1$    | $P_4$                     |





What is the proper expression for the cycle efficiency?

A.  $\frac{{}_3W_4}{{}_2Q_3}$

B.  $\frac{{}_1W_2 + {}_3W_4}{{}_2Q_3}$

C.  $\frac{{}_1W_2 + {}_3W_4}{{}_2Q_3 + {}_3Q_4}$  ✓ ✓

$$\eta_{\text{th}} = \frac{W_{\text{net}}}{Q_{\text{in}}}$$

# Solving for heat and work

$$\underline{\bar{W}_2} = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{MRT}{V} dV = MRT_1 \ln\left(\frac{V_2}{V_1}\right), \quad {}_3\bar{W}_4 = MKT_3 \ln\left(\frac{V_1}{V_2}\right)$$

$$\Rightarrow \Delta KE = \Delta PE = 0 \rightarrow \Delta \bar{U} = {}_1Q_2 - {}_1\bar{W}_2$$

$${}_1Q_2 = \underbrace{\Delta \bar{U}} + {}_1\bar{W}_2$$

$$\Delta \bar{U} = M \Delta u$$
$$= M C_v \Delta T$$

ideal gas  
 $C_v = \text{const}$